Line Plots

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```
library(arrow)
```

```
##
## Attaching package: 'arrow'

## The following object is masked from 'package:utils':
##
##     timestamp
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(ggplot2)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following object is masked from 'package:arrow':
##
##     duration

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library(scales)

# 1. Define paths based on your Analyzing_Unemployment folder on the Desktop
DATA_DIR     <- "~/Desktop/Analyzing_Unemployment"
PARQUET_PATH <- file.path(DATA_DIR, "cps_master_panel_2005_2026.parquet")
HAZARD_CSV   <- file.path(DATA_DIR, "cps_monthly_hazards.csv")

# 2. Load Parquet Data (efficient column selection to prevent crashes)
cols_to_load <- c("YEAR", "MONTH", "CPSIDP", "EMPSTAT", "IND", "WTFINL", "MISH")

df <- read_parquet(PARQUET_PATH, col_select = all_of(cols_to_load)) %>%
  filter(EMPSTAT %in% c(10, 11, 12, 20, 21))

print(paste("Successfully loaded rows:", nrow(df)))
```

```
## [1] "Successfully loaded rows: 7096501"
```

```r
# 2. Data Cleaning & Longitudinal Linking for Transition Rates
df <- df %>%
  mutate(
    Date = make_date(year = YEAR, month = MONTH, day = 1),
    Employed = ifelse(EMPSTAT %in% c(10, 11, 12), 1, 0),
    Unemployed = ifelse(EMPSTAT %in% c(20, 21), 1, 0)
  )

# Group major industries based on CPS industry codes
df <- df %>%
  mutate(Industry_Group = case_when(
    IND >= 170 & IND <= 290  ~ "Agriculture & Mining",
    IND >= 370 & IND <= 499  ~ "Construction",
    IND >= 1070 & IND <= 3990 ~ "Manufacturing",
    IND >= 4070 & IND <= 4590 ~ "Trade, Transportation & Utilities",
    IND >= 5070 & IND <= 5790 ~ "Information & Financial Activities",
    IND >= 6070 & IND <= 6390 ~ "Professional & Business Services",
    IND >= 6470 & IND <= 7690 ~ "Education & Health Services",
    IND >= 7770 & IND <= 8470 ~ "Leisure & Hospitality",
    TRUE ~ "Other Services"
  ))

# Sort by person identifier and time to track consecutive months
df <- df %>% arrange(CPSIDP, YEAR, MONTH)

# Calculate month-over-month transitions using lag within each individual (CPSIDP)
df_transitions <- df %>%
  group_by(CPSIDP) %>%
  mutate(
    Prev_Employed = lag(Employed),
    Prev_Unemployed = lag(Unemployed),
    Prev_Date = lag(Date),
    Prev_Ind = lag(Industry_Group)
  ) %>%
  ungroup() %>%
  filter(!is.na(Prev_Employed) & Date == (Prev_Date + months(1)))
```

```r
# Calculate monthly Separation Rate and Job Finding Rate by Industry
transition_rates <- df_transitions %>%
  filter(Industry_Group == Prev_Ind) %>%
  group_by(Date, Industry_Group) %>%
  summarize(
    At_Risk_E = sum(Prev_Employed == 1, na.rm = TRUE),
    Separations = sum(Prev_Employed == 1 & Unemployed == 1, na.rm = TRUE),
    Separation_Rate = Separations / At_Risk_E,

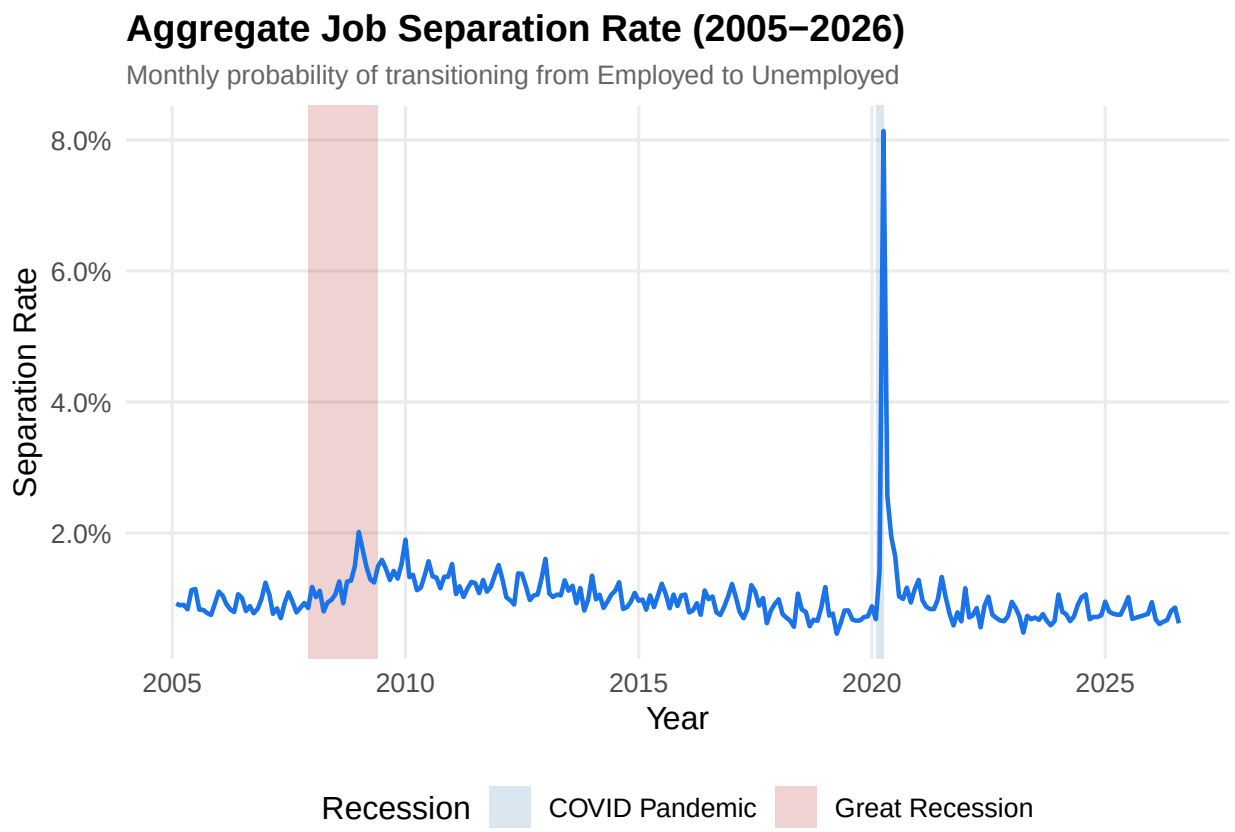
    At_Risk_U = sum(Prev_Unemployed == 1, na.rm = TRUE),
    Findings = sum(Prev_Unemployed == 1 & Employed == 1, na.rm = TRUE),
    Finding_Rate = Findings / At_Risk_U,
    .groups = "drop"
  )

# Define Recession Shading Rectangles & Plot Theme
recessions <- data.frame(
  start = as.Date(c("2007-12-01", "2020-02-01")),
  end = as.Date(c("2009-06-01", "2020-04-01")),
  Event = c("Great Recession", "COVID Pandemic")
)

theme_dsdc <- theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(color = "gray40", size = 10),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
```

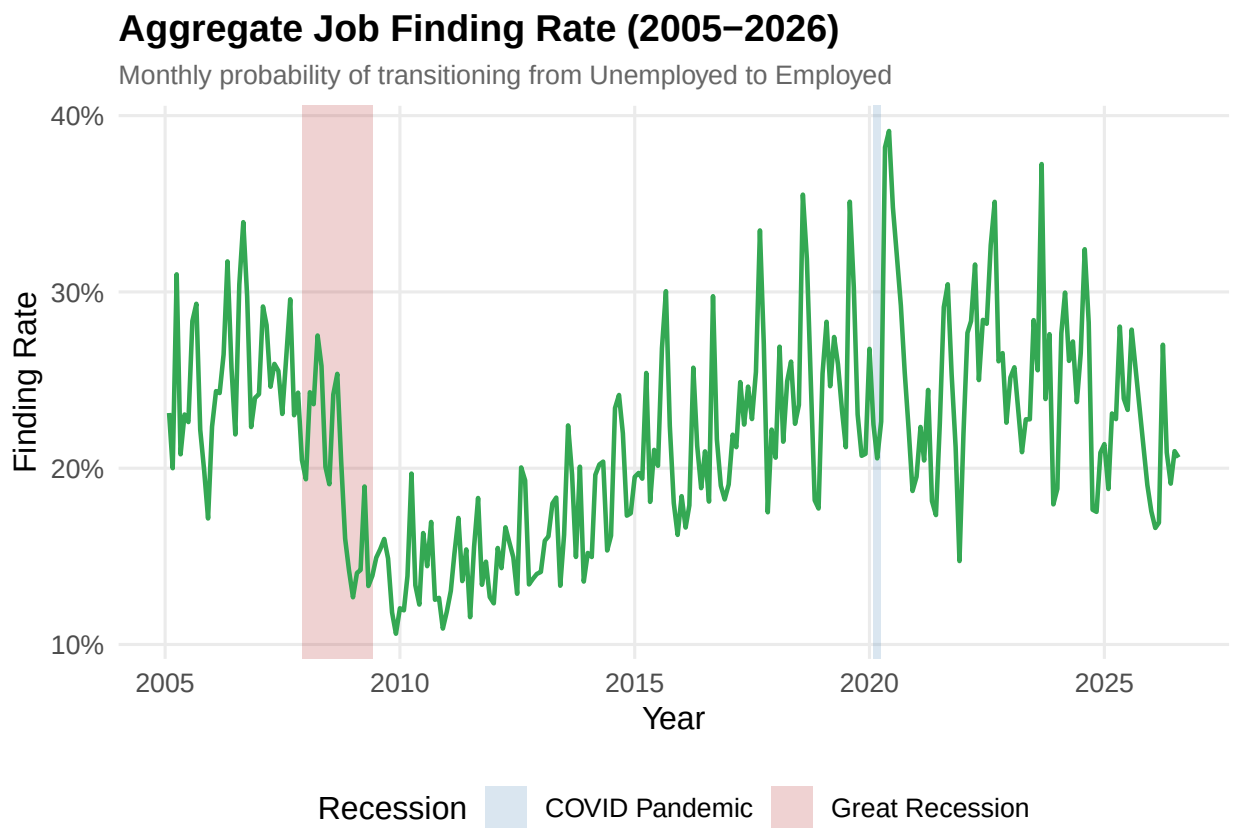
```r
# --- Plot 1: Overall Monthly Separation Rate Across Time ---
overall_sep <- transition_rates %>%
  group_by(Date) %>%
  summarize(Separation_Rate = sum(Separations) / sum(At_Risk_E))

p1 <- ggplot(overall_sep, aes(x = Date, y = Separation_Rate)) +
  geom_rect(data = recessions, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = Event),
            inherit.aes = FALSE, alpha = 0.2) +
  scale_fill_manual(values = c("Great Recession" = "firebrick", "COVID Pandemic" = "steelblue")) +
  geom_line(color = "#1a73e8", linewidth = 0.8) +
  scale_y_continuous(labels = percent_format(accuracy = 0.1)) +
  labs(
    title = "Aggregate Job Separation Rate (2005-2026)",
    subtitle = "Monthly probability of transitioning from Employed to Unemployed",
    x = "Year", y = "Separation Rate", fill = "Recession"
  ) +
  theme_dsdc
print(p1)
```

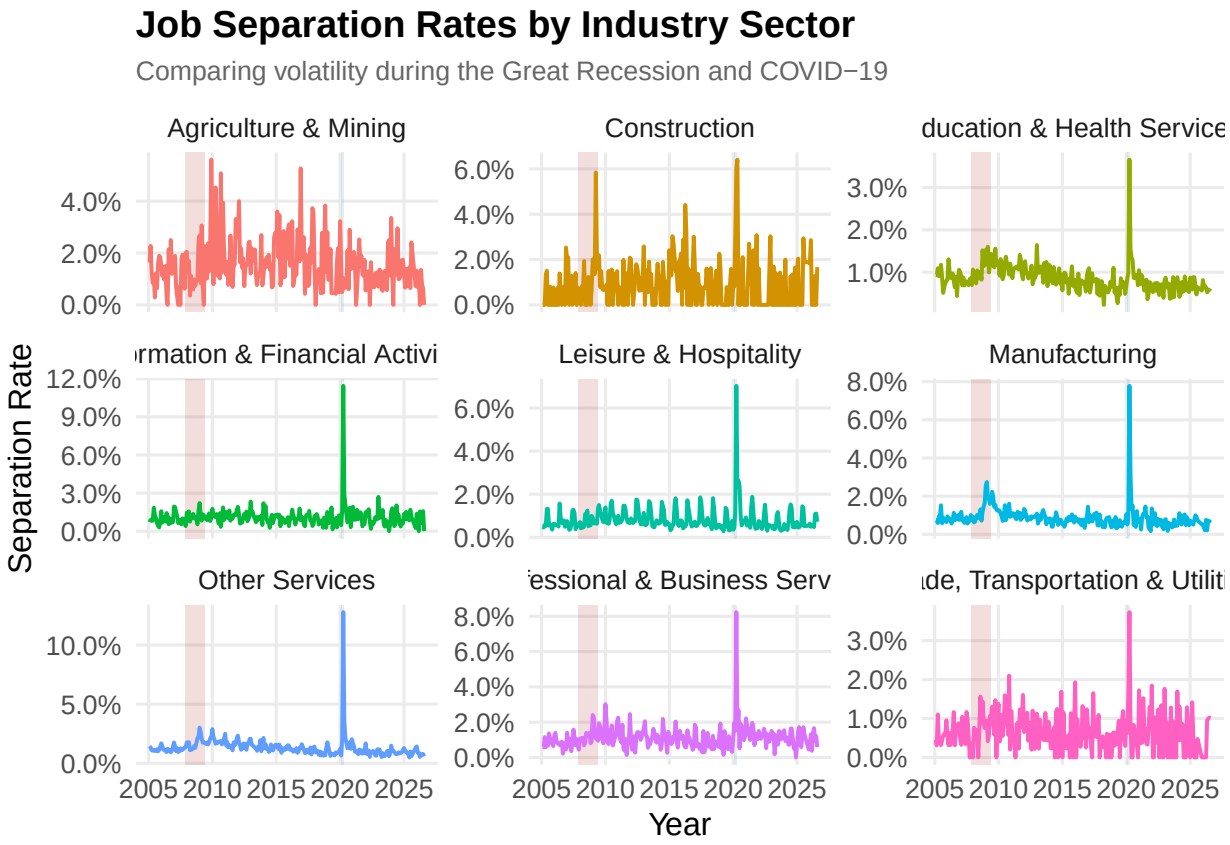



```
# --- Plot 2: Overall Monthly Job Finding Rate Across Time ---
overall_find <- transition_rates %>%
  group_by(Date) %>%
  summarize(Finding_Rate = sum(Findings) / sum(At_Risk_U))

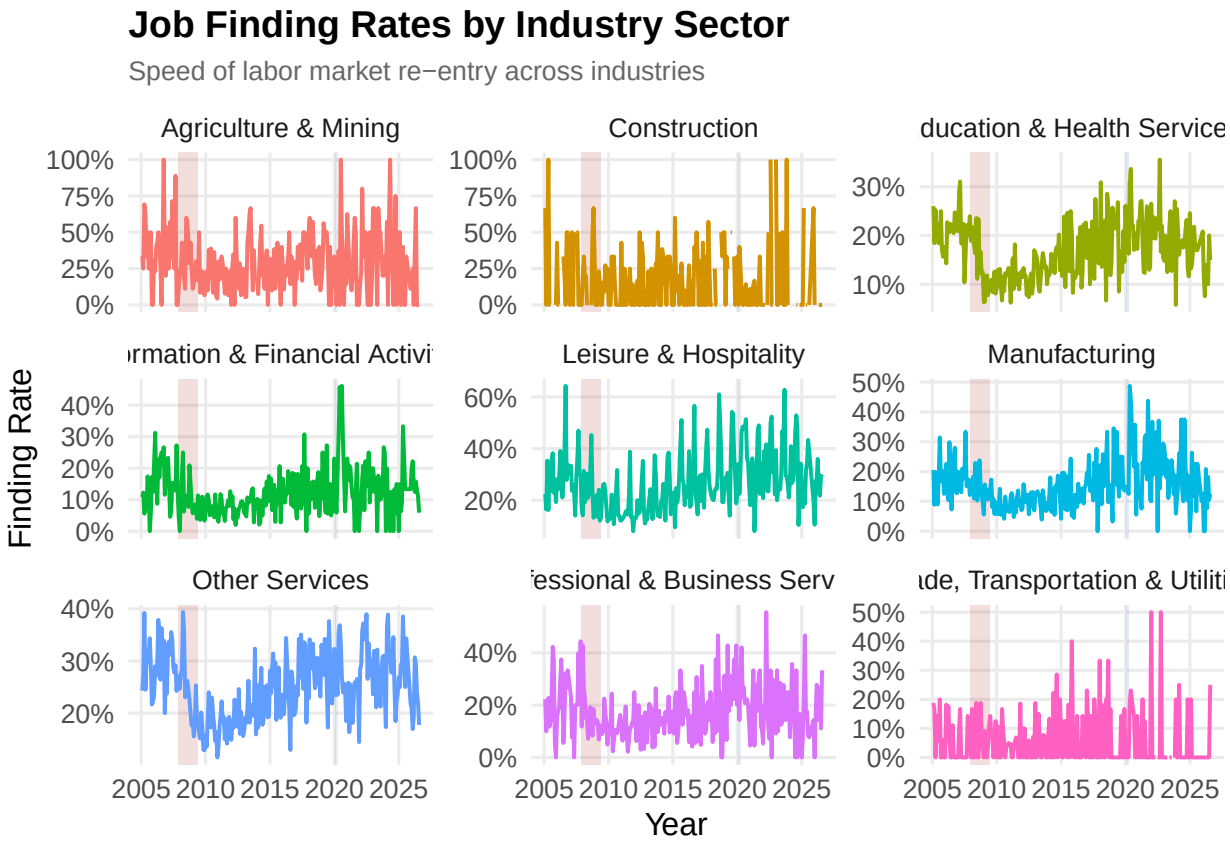
p2 <- ggplot(overall_find, aes(x = Date, y = Finding_Rate)) +
  geom_rect(data = recessions, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = Event),
            inherit.aes = FALSE, alpha = 0.2) +
  scale_fill_manual(values = c("Great Recession" = "firebrick", "COVID Pandemic" = "steelblue")) +
  geom_line(color = "#34a853", linewidth = 0.8) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    title = "Aggregate Job Finding Rate (2005-2026)",
    subtitle = "Monthly probability of transitioning from Unemployed to Employed",
    x = "Year", y = "Finding Rate", fill = "Recession"
  ) +
  theme_dsdc
print(p2)
```




```r
# --- Plot 3: Industry-Specific Separation Rates (Faceted Line Plot) ---
p3 <- ggplot(transition_rates, aes(x = Date, y = Separation_Rate, color = Industry_Group)) +
  geom_rect(data = recessions, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = Event),
            inherit.aes = FALSE, alpha = 0.15) +
  scale_fill_manual(values = c("Great Recession" = "firebrick", "COVID Pandemic" = "steelblue")) +
  geom_line(linewidth = 0.7) +
  facet_wrap(~ Industry_Group, scales = "free_y", ncol = 3) +
  scale_y_continuous(labels = percent_format(accuracy = 0.1)) +
  labs(
    title = "Job Separation Rates by Industry Sector",
    subtitle = "Comparing volatility during the Great Recession and COVID-19",
    x = "Year", y = "Separation Rate", fill = "Recession"
  ) +
  theme_dsdc +
  theme(legend.position = "none")
print(p3)
```




```r
# --- Plot 4: Industry-Specific Job Finding Rates (Faceted Line Plot) ---
p4 <- ggplot(transition_rates, aes(x = Date, y = Finding_Rate, color = Industry_Group)) +
  geom_rect(data = recessions, aes(xmin = start, xmax = end, ymin = -Inf, ymax = Inf, fill = Event),
            inherit.aes = FALSE, alpha = 0.15) +
  scale_fill_manual(values = c("Great Recession" = "firebrick", "COVID Pandemic" = "steelblue")) +
  geom_line(linewidth = 0.7) +
  facet_wrap(~ Industry_Group, scales = "free_y", ncol = 3) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    title = "Job Finding Rates by Industry Sector",
    subtitle = "Speed of labor market re-entry across industries",
    x = "Year", y = "Finding Rate", fill = "Recession"
  ) +
  theme_dsdc +
  theme(legend.position = "none")
print(p4)
```


Job Finding Rates by Industry Sector
Speed of labor market re−entry across industries
Agriculture & Mining
Construction
ducation & Health Service
rmation & Financial Activi
Leisure & Hospitality
Manufacturing
Other Services
essional & Business Serv
ıde, Transportation & Utiliti
Finding Rate
Year
2005 2010 2015 2020 2025